



Final Round 2025

First Exam – Solutions

Duration: 4 hours

Difficulty: The problems are ordered by difficulty.

Points: Each problem is worth 7 points.

Aarburg

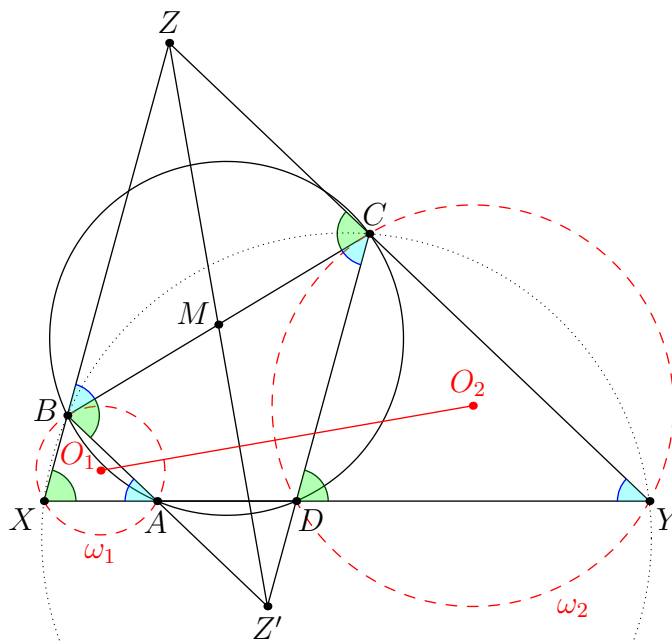
March 21, 2025

Preliminary Remark

A complete solution is worth 7 points. For every problem, up to 2 points can be deducted from a full solution for (minor) flaws. Partial marks are attributed according to the marking schemes. In the case of multiple marking schemes for the same problem, the score is the maximum among all the marking schemes.

1. Let $ABCD$ be a cyclic quadrilateral without parallel sides. Let points X and Y lie on DA such that $BX \parallel CD$ and $CY \parallel AB$. Let Z be the intersection of BX and CY , and M be the midpoint of the segment BC .

Solution



We will prove that MZ is the radical axis of the circumcircles ω_1 and ω_2 of ABX and CDY respectively. In particular, this would imply that MZ is perpendicular to the line connecting the circumcenters, as desired. We start by proving that $BCXY$ is cyclic. This can be done by angle chasing as

$$\angle BXY + \angle YCB = \angle DCB + \angle BAD = 180^\circ$$

by the parallelisms, as $ABCD$ is cyclic. Therefore, as $Z = BX \cap CY$, we can compute the power of Z at $(BCXY)$: $ZX \cdot ZB = ZY \cdot ZC$. That means that Z has the same power with respect to ω_1 and ω_2 : it lies on the radical axis.

Now, we claim that BC is tangent to ω_1 : again, by angle chasing,

$$\angle CBX = \angle CYD = \angle BAX$$

by the above cyclicity and parallelism. Similarly, BC is tangent to ω_2 . Therefore the power of M with respect to ω_1 , respectively ω_2 is MB^2 , respectively MC^2 . As $MB = MC$, it follows that these powers are equal, which proves that M also lies on the radical axis. This concludes the solution.

Solution 2: We introduce Z' the intersection of AB and CD . By parallelism, $Z'BZC$ is a parallelogram, and therefore Z, Z' and M are collinear. Therefore, with the same notation as in Solution 1, if Z' lies on the radical axis of ω_1 and ω_2 we can conclude the same way as above (ZZ' will be the desired radical axis).

But, by computing the power of Z' with respect to $(ABCD)$ we get that $Z'B \cdot Z'A = Z'C \cdot Z'D$, which is enough to conclude that Z' has the same power with respect to ω_1 and ω_2 .

Marking Scheme

Among (A), (B) and (C), the student gets awarded the sum of their best two scores. Each section is additive and (D) is additive to the rest.

(A) **3P:** Working with Z

(A1) **2P:** Showing that $BCXY$ is cyclic

(A2) **1P:** Deducing that $\text{Pow}_{\omega_1}(Z) = \text{Pow}_{\omega_2}(Z)$

(B) **3P:** Working with M

(B1) **2P:** Showing that BC is tangent to ω_1 and ω_2

(B2) **1P:** Deducing that $\text{Pow}_{\omega_1}(M) = \text{Pow}_{\omega_2}(M)$

(C) **3P:** Working with Z'

(C1) **1P:** Introducing Z'

(C2) **1P:** Justifying that Z, Z' and M are collinear

(C3) **1P:** Showing that $\text{Pow}_{\omega_1}(Z') = \text{Pow}_{\omega_2}(Z')$

(D) **1P:** Concluding

If the student claims that MZ is the radical axis of ω_1 and ω_2 without proving it and if no other points are awarded, they get **1P**.

2. Determine all functions $f: \mathbb{R}_{>0} \rightarrow \mathbb{R}_{>0}$ such that

$$y \cdot \min\left(f(xy), f(x)\right) = \min\left(f\left(\frac{x}{y}\right), f(x)\right)$$

for all $x, y \in \mathbb{R}_{>0}$.

Solution

First of all, let $x > 0, y > 1$ real numbers. We have the chain of inequalities

$$f(x) \geq \min\left(f\left(\frac{x}{y}\right), f(x)\right) = y \cdot \min(f(xy), f(x)).$$

If $\min(f(xy), f(x)) = f(x)$, this would imply $f(x) \geq yf(x)$, which is absurd as $y > 1$ and $f(x) > 0$. So $\min(f(xy), f(x)) = f(xy)$ and we have proven that

$$f(x) \geq yf(xy) > f(xy)$$

where the second inequality comes from the fact that $y > 1$. In particular, for any real numbers $0 < a < b$, we can set $x := a > 0, y := b/a > 1$ and this implies that $f(a) = f(x) > f(xy) = f(b)$, which proves that f is decreasing.

Now, if we take $x > 0, y < 1$ real numbers, then $xy < x$ and $\frac{x}{y} > x$ so $\min(f(xy), f(x)) = f(x)$ and $\min\left(f\left(\frac{x}{y}\right), f(x)\right) = f\left(\frac{x}{y}\right)$ as f is decreasing. This implies that $yf(x) = f\left(\frac{x}{y}\right)$, in particular by letting $x = y$ we get that $yf(y) = f(1)$ for $y < 1$. This is trivially true if $y = 1$, and if $y > 1$, we get $xy > x$ and $\frac{x}{y} < x$ so $\min(f(xy), f(x)) = f(xy)$ and $\min\left(f\left(\frac{x}{y}\right), f(x)\right) = f(x)$ as f is decreasing. This implies that $yf(xy) = f(x)$, in particular by letting $x = 1$ we get that $yf(y) = f(1)$ for $y > 1$.

So $yf(y) = f(1) \iff f(y) = \frac{f(1)}{y}$ for all $y > 0$ and one can easily check that for any real number $c > 0$, the function $f(x) = \frac{c}{x}$ is a solution of the functional equation.

In conclusion, all the functions satisfying the functional equation are of the form $\frac{c}{x}$, where $c > 0$ is a constant real number.

Marking Scheme

- (A) **1P:** Checking that all functions of the form $f(x) = c/x$ with $c \in \mathbb{R}_{>0}$ work.
- (B) **3P:** Showing that f is decreasing.
 - (B1) **1P:** Showing that $yf(x) \geq \min\left(f\left(\frac{x}{y}\right), f(x)\right)$ for any $x, y > 0$.
 - (B2) **1P:** Showing that $f(x) \geq y \cdot \min(f(xy), f(x))$ for any $x, y > 0$.
 - (B3) **2P:** Showing that $yf(x) \geq f\left(\frac{x}{y}\right)$ when $y < 1$.
 - (B4) **2P:** Showing that $f(x) \geq yf(xy)$ when $y > 1$.
- (C) **1P:** Showing that for $y > 1$, $yf(y) = f(1)$.
- (D) **1P:** Showing that for $y < 1$, $yf(y) = f(1)$.

Section B is non-additive.

Items B1 and B2 are also awarded if the contestant assumes one of $y > 1$ or $y < 1$.

3. Let n be a positive integer. A group of n penguins swim in n races, in each of which they are ranked from 1st to n th position with no draws. A penguin is eligible for a rating (a, b) for positive integers a and b , if in at least a races they finished in any of the first b positions. Their final score is the maximum possible value $a - b$ across all ratings for which they are eligible.

Find the maximum possible sum of all scores of the n penguins.

Solution

Answer: The greatest possible value that can be achieved is $\frac{n(n-1)}{2}$.

Construction: $\frac{n(n-1)}{2}$ can be achieved by having P_i finish i^{th} in every race. Since P_i ranked in n races among the first i penguins, $(a_i - b_i)$ can be $(n - i)$. The following summation gives us the answer:

$$\sum_{i=1}^n n - i = \frac{n(n-1)}{2}$$

Solutions 1 & 2 describe an algorithm which modifies current rankings to the ranking given in the construction above, without reducing the total score and thus showing that the construction gives the greatest possible value.

Solutions 3 & 4 prove the bound via double counting.

Solution 5 assigns weights to every position in every race and bounds the score to the total weight of all the positions.

Solution 1: To show the maximum, we will apply a series of modifications to the results of the races, each of which does not decrease the total score, such that after k such modifications the first k positions are the same in every race. Say the score of P_i is the maximum possible value of $(a_i - b_i)$. Say that P_i is scored on the b^{th} place if their score is $a - b$ because they came in the top b places in a races and b is the minimal with this property for P_i .

We will now show that there exists a modification to make at least one penguin be scored on the k^{th} place. Assuming that the first $k - 1$ positions are the same in every race and that there is not a penguin scored on the k^{th} place. Let $l > k$ be the minimal such that P_s is scored on the l^{th} place. Now in every race where P_s ranks in the positions $[k, l]$, reorder the penguins such that P_s finishes in the k^{th} rank. Since every other penguin is scored on a position l or lower, swapping positions in the range of $[k, l]$ will not decrease their score and the score of P_s can only increase as its positions in every race can only increase. After this change P_s can be scored on the k^{th} position, which not decrease the total score as P_s no longer ranks $[k + 1, l]$ in any race.

Now we know that at least one penguin is scored on the k^{th} position. Pick a penguin P_s that is ranked on the k^{th} position. In every race where P_s does not finish in the k^{th} position, swap it with P_t , who finished in the k^{th} position. Notice that P_t 's score can decrease by at most one if $a_t > 1$ and if $a_t = 1$ then taking $a_t = n$ also ensures that the score will decrease by at most 1. P_s 's score will increase by exactly one as it is scored on the k^{th} position and it gains an additional placement in the top k positions. Therefore each swap does not decrease the total score. At the end the first k positions will be the same in every race.

Repeating this n times leads to the construction given above and thus shows that any position will have a value of at most $\frac{n(n-1)}{2}$.

Solution 2: Consider P_s such that P_s did not finish in the same position in every race. This is possible, unless we have the construction above. We will now swap pairs of penguins until P_s finished in the same position in every race, namely position b_s . Assume that in race x , P_s did not rank in position b_s but position p_s . Let P_t be the penguin that ranked in the b_s position in race x .

- If $p_s > b_s$, then swapping P_s 's and P_t 's positions in race x will keep P_s 's score the same and will not decrease P_t 's score.
- If $p_s < b_s$, then swapping P_s 's and P_t 's positions in race x will increase P_s 's score by 1 and decrease P_t by at most 1. If was $a_t > 1$ then clearly the score can be reduced by at most 1 by reducing a_t by 1 if needed. If $a_t = 1$ then the original score was $1 - b_t \leq 1 - 1 = 0$ and a non negative score is always possible by taking $a_n = n$ and $b_n = n$.

Therefore swapping until P_s ranked in the same position (b_s) in all of the races does not decrease the total score. There will always exist a P_s with b_s not being a position that has a penguin that always ranks in that position, unless we reach our construction. By repeating this for every penguin we reach our construction above and thus show that any position will have a value of at most $\frac{n(n-1)}{2}$.

Solution 3: Let us count the number of occurrences of a penguin beating another penguin in n races. In each race there are $\frac{n(n-1)}{2}$ pairs of penguins beating another penguin and thus $n\frac{n(n-1)}{2}$ is the total number of occurrences over all the races.

Now let us count the number of times a penguin (P_i) beats another penguin and contributes to his score with it. In a single race where P_i finishes in the top b_i positions, they must have beaten at least $n - b_i$ other penguins. Doing this for all the races where P_i finished in the top b_i positions gives us $a_i(n - b_i)$. Summing up over all penguins gives us:

$$\sum_{i=1}^n a_i(n - b_i) \leq n\frac{n(n-1)}{2}$$

Notice that $n(a_i - b_i) \leq a_i(n - b_i)$ as $a_i \leq n$. Therefore we have:

$$n \sum_{i=1}^n (a_i - b_i) \leq \sum_{i=1}^n a_i(n - b_i) \leq n\frac{n(n-1)}{2}$$

Dividing by n on both sides leads to the required bound, which can be obtained with the construction above.

Solution 4: Let the score of a penguin P_i be the maximum value of $a_i - b_i$. Consider a penguin whose rankings in the n races were $r_1, r_2, \dots, r_n$ and score s . We will show $\sum r_i \leq n(n - s)$.

Without loss of generality let $r_1 \leq r_2 \leq \dots \leq r_n$. There must exist some k such that $s+1 \leq k \leq n$ and $k - r_k = s$ (k would be a_i and r_k would be b_i). We can maximize $\sum r_i$ by setting every $r_1, r_2, \dots, r_{k-1}$ to r_k and setting $r_{k+1}, \dots, r_n$ to n .

$$\sum r_i \leq kr_k + (n - k)n = n^2 - k(n - r_k) = n^2 - k(n + s - k) \leq n^2 - ns = n(n - s)$$

The final inequality follows as $k(n + s - k)$ is minimized when $k = n$ given that $s + 1 \leq k \leq n$.

The sum of the ranks over all the penguin is $n\frac{n(n+1)}{2}$ and setting t as the total score of all penguins gives us.

$$\frac{n^2(n+1)}{2} \leq n^3 - tn$$

Via rearrangements we get $t \leq \frac{n(n-1)}{2}$ which was desired.

Solution 5: Let the score of a penguin P_i be the maximum possible value of $a_i - b_i$. In each race, assign the penguin in the k^{th} position a weight of $1 - \frac{k}{n}$. Consider a penguin P_i who finished in the top b_i places in at least a_i races. P_i 's total weight is at least $a(1 - \frac{b}{n}) = a - b(\frac{a}{n}) \geq a - b$. Therefore the score of every penguin is at most the total weight across all races of that penguin. The sum of weights in each race is $\frac{n-1}{2}$ and therefore over all the races it is $\frac{n(n-1)}{2}$. Therefore the total score is at most $\frac{n(n-1)}{2}$ which can be achieved by the construction above.

Marking Scheme

The following marking schemes are additive. **Solution 1:**

- (A) **1P:** Proving the lower bound $\frac{n(n-1)}{2}$.
- (B) **1P:** Attempt to design an algorithm that changes the placement of penguins while not decreasing the total score.
- (C) **4P:** Provide a valid algorithm
 - (C1) **1P:** Assume that the first $k - 1$ positions are the same in every race and attempt to show that you can make the k^{th} position the same in every race.
 - (C2) **1P:** Show that this is possible if there exists a penguin with $b_i = k$.
 - (C3) **2P:** Show that one can always get a penguin with $b_i = k$.

Solution 2:

- (A) **1P:** Proving the lower bound $\frac{n(n-1)}{2}$.
- (B) **1P:** Attempt to design an algorithm that changes the placement of penguins while not decreasing the total score.
- (C) **4P:** Provide a valid algorithm
 - (C1) **1P:** Show that after every swap of two positions each penguin's score can go down by at most 1 (in particular if $a_i = 0$).
 - (C2) **2P:** Show that if for some penguins P_i and P_j , such that P_j finished in the b_i position, we can swap a non b_i position from P_i for b_i position from P_j and not decrease the total score.
 - (C3) **-1P:** Forget to state that this algorithm can terminate only if all penguins have the same rank in every race.

Solution 3:

- (A) **1P:** Proving the lower bound $\frac{n(n-1)}{2}$.
- (B) **1P:** Show that there are $\frac{n^2(n-1)}{2}$ occurrences of a penguin beating another penguin in any race.
- (C) **1P:** Express the number of beatings as the sum of the number of penguins P_i beats.
- (D) **2P:** Bound the total number of beatings by the number of times a penguin P_i beats another penguin P_j , over all races where P_i is ranked among the top b_i .

Remark: If a student went from (B) to (D) without writing (C), they still get the (C) point.

Solution 4:

- (A) **1P:** Proving the lower bound $\frac{n(n-1)}{2}$.
- (B) **1P:** Show that the sum of all ranks over all races is $n\frac{n(n+1)}{2}$.
- (C) **1P:** Express the total sum of ranks as the sum of ranks of penguin P_i .
- (D) **1P:** Stating that the values of the ranks can be reduced to two values b_i and n . Only applicable if the sum of ranks is considered.
- (E) **1P:** Bound the sum of ranks for a specific penguin to n and $(a_i - b_i)$ in a useful way.
- (F) **1P:** Setup an inequality that leads to a finish via algebraic manipulations.

Solution 5: The marking scheme for solution 3 is used. If a contestant gets less than 2 points they can get an additional point for assigning useful weights to every position.

4. Determine all infinite sequences $a_1, a_2, \dots$ of positive integers such that, for any integer $n \geq 2$, both the arithmetic and geometric mean of any n consecutive terms of the sequence are integers.

Remark: Integers $x_1, \dots, x_k > 0$ have arithmetic mean $\frac{x_1 + \dots + x_k}{k}$ and geometric mean $\sqrt[k]{x_1 \cdots x_k}$.

Solution

Answer: The only sequences that work are the constant sequences.

Lemma 1. The sequence $a_1, a_2, \dots \pmod{n}$ has period n for all $n \in \mathbb{N}$.

Proof. Consider any n consecutive terms $a_i, a_{i+1}, \dots, a_{i+n-1}$. Since the arithmetic mean of these terms is an integer, their sum must be divisible by n . But this is also true for the n consecutive terms $a_{i+1}, a_{i+2}, \dots, a_{i+n}$. Thus, $a_i \equiv a_{i+n} \pmod{n}$ for all $i \geq 1$. $\square$

Lemma 2. The sequence $v_p(a_1), v_p(a_2), \dots \pmod{n}$ has period n for any $n \in \mathbb{N}$, p prime.

Proof. Consider any n consecutive terms $a_i, a_{i+1}, \dots, a_{i+n-1}$. Since the geometric mean of these terms is an integer, their product must be an n -th power. In particular, the exponent of p in this product must be divisible by n . But this is also true for the n consecutive terms $a_{i+1}, a_{i+2}, \dots, a_{i+n}$ and thus $v_p(a_i) \equiv v_p(a_{i+n}) \pmod{n}$ for all $i \geq 1$. $\square$

Since constant sequences clearly work, let us assume the sequence is not constant. This means there is a prime p such that $v_p(a_i)$ is not constant. In particular, there are terms a_x and a_y with

$$\alpha = v_p(a_x) < v_p(a_y) = \beta.$$

Now pick some integer $k > v_p(a_y)$ that is not divisible by p .

By lemma 1, the sequence $a_i \pmod{p^\alpha}$ is periodic with period p^α . By lemma 2, the sequence $v_p(a_i) \pmod{k}$ is periodic with period k . Since $p \nmid k$, by the Chinese remainder theorem we can find an index $z \geq 1$ such that $p^\beta \mid z - x$ and $k \mid z - y$. Thus, $a_z \equiv a_x \equiv p^\alpha \not\equiv 0 \pmod{p^\beta}$ and $v_p(a_z) \equiv v_p(a_y) \equiv \beta \pmod{k}$. But since $k > v_p(a_y)$, this implies $v_p(a_z) \geq \beta$. But the first congruence implies $p^\beta \nmid a_z$ while the second implies $p^\beta \mid a_z$, a contradiction. We conclude that the only sequences that work are the constant sequences we excluded in the beginning.

Marking Scheme

- (A) **1P:** Claiming the answer is exactly the constant sequences
- (B) **1P:** Periodicity of $a_i \pmod n$ for sufficiently general n
- (C) **2P:** Periodicity of $v_p(a_i) \pmod n$ for sufficiently general (independent) p and n
- (D) **1P:** Arguing that if a_i is not constant, then $v_p(a_x) \neq v_p(a_y)$ for some x, y, p (or similar)
- (E) **1P:** Using the Chinese Remainder Theorem on two coprime periods (in a meaningful way)

Item A is only awarded if no other points are obtained.

The fact that a constant sequences work can be stated without proof.

At most 5 points are awarded to contestants who mistakenly assume the mean condition is invariant under scaling the sequence.



Final Round 2025

Second Exam – Solutions

Duration: 4 hours

Difficulty: The problems are ordered by difficulty.

Points: Each problem is worth 7 points.

Aarburg

March 22, 2025

Preliminary Remark

A complete solution is worth 7 points. For every problem, up to 2 points can be deducted from a full solution for (minor) flaws. Partial marks are attributed according to the marking schemes. In the case of multiple marking schemes for the same problem, the score is the maximum among all the marking schemes.

5. Determine all triples of positive integers (p, q, a) such that p and q are prime numbers and

$$p^q - q^a = 2025.$$

Solution

Answer: There are no such triples.

If both p and q are odd, $p^q - q^a$ is even which is a contradiction to $p^q - q^a = 2025$. Therefore, either p or q must be even. Since p and q are primes, this leaves us with 2 cases:

- (a) $p = 2$: In this case we have

$$2^q - q^a = 2025.$$

By Fermat's little Theorem:

$$2^q - q^a \equiv 2^q \equiv 2 \equiv 2025 \pmod{q},$$

which yields $2 \equiv 2025 \pmod{q}$, implying $2023 \equiv 0 \pmod{q}$, so $q \mid 2023 = 7 \cdot 17^2$. This leaves the cases $q = 7$ and $q = 17$. For $q \geq 2$ one can see that $2^q - q^a \equiv -q^a \equiv 2025 \equiv 1 \pmod{4}$, which does not have a solution for $q \equiv 1 \pmod{4}$. This only leaves the case $q = 7$, which does not lead to a solution, as $2^7 < 2025$.

- (b) $q = 2$: We have $p^2 - 2^a = 2025 \iff p^2 - 2025 = 2^a$. Knowing that $2025 = 45^2$ we realize that the left side of the equation can be factorized:

$$(p - 45)(p + 45) = 2^a$$

Therefore, both $p - 45$ and $p + 45$ are powers of 2. Let $p + 45 = 2^b$, then $p - 45 = 2^b - 90$ is also a power of 2. Note that for any $2^b \geq 256$, $2^b - 90 > 2^{b-1}$ so $2^b \leq 128$. Furthermore, $2^b - 90 < 0$ for $2^b \leq 64$, therefore the only possible value for $2^b = 128$. But $128 - 90 = 38$ is not a power of 2, so there are no solutions in this case.

Marking Scheme

(A) **1P:** Showing that $p = 2$ or $q = 2$

(B) **1P:** Looking at the equation mod q and applying Euler-Fermat.

(C) **2P:** Showing that the case $p = 2$ has no solution:

The following points are additive:

(C1) **1P:** Concluding from Euler-Fermat that $2023 \equiv 0 \pmod{q}$.

(C2) **1P:** Showing that there is no solution for $q \mid 2023$.

(D) **3P:** Showing that the case $q = 2$ has no solution:

The following points are additive:

(D1) **1P:** $(p - 45)(p + 45) = 2^a$ and concluding that both $p - 45$ and $p + 45$ are powers of 2.

(D2) **1P:** Arguing that a is bounded.

(D3) **1P:** Checking the remaining cases.

6. Let n, a, b be positive integers with $n \geq 2$. Aru and Wero are playing a game on a $n \times n$ grid. At the beginning, there is a single pebble in the bottom-left corner. A move consists of removing one pebble from some non-empty square S and performing at least one (possibly both) of:
- adding a pebbles to the right neighbour of S ;
 - adding b pebbles to the upper neighbour of S .

Aru starts and they alternate moves. The first player that cannot make a move loses. Determine who, if anyone, has a winning strategy in terms of n, a, b .

Solution

We prove that Aru always wins. Note that Aru is always able to make an initial move. We will prove that with a well-chosen initial move, she will be able to "reply" to every subsequent move by Wero in such a manner that Wero can never deny her the ability to move.

Firstly, we briefly prove that the game always ends. Consider the diagonals of the grid (in the top-left to bottom-right) orientation; the first diagonal consists of the bottom-left square, the second one consists of its two neighbours, and so forth up until the $2n - 1$ th diagonal, which is the top-right square. We will induct on these diagonals to show that at most finitely many moves can be performed removing a square from each diagonal.

For the base case, it's trivial that we can only make one move removing a pebble from the first diagonal. Suppose now only finitely many moves can be performed removing from the k th diagonal. These are the only moves that introduce pebbles to the $(k + 1)$ th diagonal, so over the course of the game, only finitely many pebbles can be placed on the $(k + 1)$ th diagonal. It follows that only finitely many moves can be made removing a pebble from the $(k + 1)$ th diagonal.

Note that the game always ending can also be proved by invariance: assign the value $(a + b)^{-2k}$ to the squares on the k th diagonal. Every move now reduces the total value of all pebbles; this value is initially 1, must remain positive and the size of the reduction is also bounded below by $(a + b)^{-2n}$, so there are at most $(a + b)^{2n}$ moves.

Imagine now that we have a copy of the board which shows the number of pebbles on each square modulo 2. Initially, there is a 1 in the bottom-left corner and 0s everywhere else. We are now going to prove that Aru can always ensure that this board is symmetric with 0s on the diagonal (with respect to the bottom-left to top-right diagonal) at the end of her turn, and that this implies that no matter what move Wero does, Aru can always make a move right after.

Firstly, we settle the small matter of her first move. If either a or b is even, Aru can simply make the move which results in adding an even number of pebbles (and removing the bottom-left pebble) and the board will contain only 0s (and hence be symmetric with 0s on diagonal). In this particular case, Aru will simply preserve the fact that the board only contains 0s henceforth. Whenever Wero makes a move, Aru will follow up with the exact same move, and this is always possible since Aru gives Wero a board with an even number of pebbles everywhere - so for every move Wero makes, Aru can replicate it.

If both are odd, Aru makes the move where she adds both a and b , and the board will contain two 1s adjacent to the bottom-left pebble, and hence still be symmetric with 0s on the diagonal.

Suppose now that we are in the case $a = b = \text{odd}$, and that Wero receives a board that is symmetric, with 0s on the diagonal. There are now a few possibilities:

- All the pebbles are in the top-right square, and Wero has lost.

- Wero makes a move removing from an odd square. Since the board was symmetric before Wero's move, Aru makes the reflection of her move (so if Wero added a pebbles, Aru adds b pebbles, but makes the move on the square that is the reflection of the square where Wero made her move). This preserves symmetry and the property that the diagonal contains only 0s.
- Wero make a move removing from an even square. The previous idea does not work now because the reflection of the even square in question may be empty. Aru can instead just copy the move to again preserve symmetry and the property that the diagonal contains only 0s.

It follows that with the above strategy, Aru always has a reply to Wero's move. Since the game is finite, Aru must eventually win.

Marking Scheme

(A) **2P: Initial partials**

- (A1) **1P:** Reducing the problem to the parity of a , b and the grid squares.
- (A2) **1P:** Proving that the game must end eventually.

(B) **2P: The case where one of a or b is even**

- (B1) **1P:** Stating that Aru should copy Wero's previous move, or any other winning strategy.
- (B2) **1P:** Completely justifying why the winning strategy can always be performed.

(C) **3P: The case where both a and b are odd**

- (C1) **2P:** Noting that Aru should reflect Wero's move if she removes from an odd square, and copy it if it removes from an even square.
- (C2) **1P:** Completely justifying why the winning strategy can always be performed.

(D) **-3P: Deductions**

- (D1) **-1P:** Neglecting to state Aru's first move in either case.
- (D2) **-1P:** Only treating a significant subset (correctors' discretion) in (B).
- (D3) **-1P:** Neglecting to state that Aru should copy Wero's move if she removes from an even square in (C).
- (D4) **-1P:** A strategy that works for (C) via symmetry that does not mention that the diagonal should be kept fully even.

7. Determine all sequences $x_1, x_2, \dots, x_n$ of rational numbers, such that for all natural numbers m ,

$$\frac{(x_1)^m + (x_2)^m + \dots + (x_n)^m}{n}$$

is equal to some rational number raised to the power of m .

Solution

Answer: Constant sequences and those of the form $(x, -x, x, \dots, -x)$ for $x \in \mathbb{Q}$ and n even (as well as their permutations).

Write

$$S_m := \frac{x_1^m + \dots + x_n^m}{n}$$

and note that if the sequence $x_1, \dots, x_n$ works, then so does the sequence

$$y_1 := \lambda x_1, \dots, y_n := \lambda x_n,$$

where λ is any positive rational number. By multiplying by their denominators' lowest common multiple, we can without loss of generality assume that $x_1, \dots, x_n$ are integers.

Let $p \mid n$ be a prime divisor of n . For S_m to be an m -th power of a rational we need to have $m \mid v_p(S_m)$. Now for $m > v_p(n)$, we have

$$v_p(S_m) \geq -v_p(n) > -m.$$

So for $v_p(S_m)$ to be divisible by m , we need $v_p(S_m) \geq 0$, implying that

$$p^{v_p(n)} \mid x_1^m + \dots + x_n^m.$$

For any $m > n \geq \max\{v_p(n) : p \mid n\}$, we now have $n \mid x_1^m + \dots + x_n^m$, implying that S_m is an integer. Let

$$A_m := \frac{|x_1|^m + \dots + |x_n|^m}{n}.$$

By the power mean inequality we have $\sqrt[m]{A_m} \leq \sqrt[m']{A_{m'}}$ for all $m \leq m'$, with equality if and only if $|x_1| = \dots = |x_n|$. The $\sqrt[m]{A_m}$ are furthermore bounded from above by $\max\{x_1, \dots, x_n\}$, and we also clearly have $A_{2k} = S_{2k}$. Assume that we don't have $|x_1| = \dots = |x_n|$. Thus,

$$\sqrt[2n]{S_{2n}} < \sqrt[2n+2]{S_{2n+2}} < \sqrt[2n+4]{S_{2n+4}} < \dots \leq C := \max\{x_1, \dots, x_n\}.$$

But a bounded sequence of strictly increasing integers can't exist; contradiction. So we must have $|x_1| = \dots = |x_n|$.

Now let a be the number of positive integers in the sequence $x_1, \dots, x_n$. Then $S_{2k} = x_1^{2k}$ and

$$S_{2k+1} = \frac{a - (n - a)}{n} x_1^{2k+1}.$$

The latter is a $(2k+1)$ -st power of a rational number if and only if $\frac{2a-n}{n}$ is a $(2k+1)$ -st power. But unless $2a - n = 0$, this clearly isn't the case for k big enough. We conclude that the only possible sequences (up to permutation) are the ones of the form $(x, x, \dots, x)$ or $(x, -x, \dots, -x)$ if n is even.

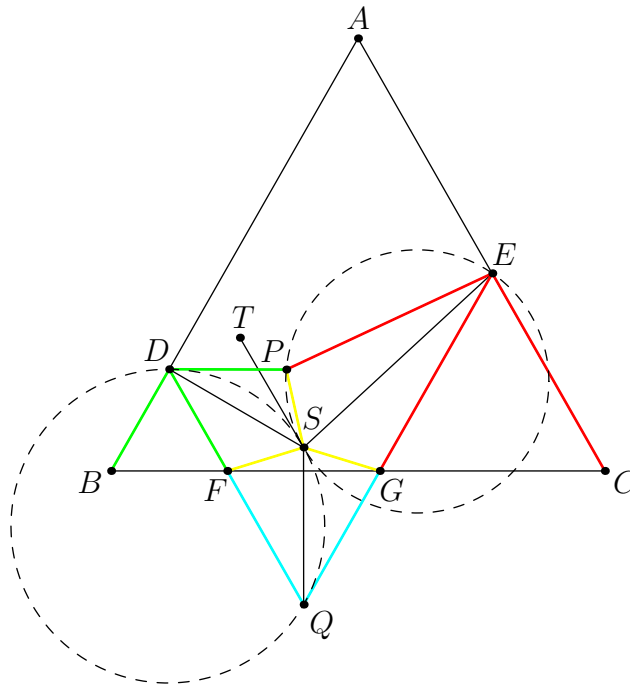
Marking Scheme

- (A) **1P:** Correctly identifying the full set of solutions.
- (B) **2P:** Proving that S_m is an integer for m big enough. (Or equivalent)
 - (B1) **1P:** Showing that for m big enough, we have $p^{v_p(n)} \mid x_1^m + \cdots + x_n^m$.
- (C) **1P:** Apply general mean inequality or AM-QM on two different $\sqrt[k]{A_k}$, with general index.
- (D) **1P:** Show that some subsequence of $\sqrt[n]{A_m}$ is bounded from above by $C = \max_{i \leq n} \{x_i\}$.
- (E) **1P:** Proving that the x_i all have the same absolute value.
- (F) **1P:** Proving that either all x_i are positive or at exactly half are.

8. Let ABC be an acute triangle with $AB = AC$. Let D and E be points on the segments AB and AC , respectively. Let ω_1 be the circle with center D and radius DB and let ω_2 be the circle with center E and radius EC . Assume that ω_1 and ω_2 intersect twice and let P be the intersection closest to BC . Denote by $F \neq B$ and $G \neq C$ the intersections of BC with ω_1 and ω_2 , respectively. Finally, let DF and EG intersect in Q , and let the angle bisectors of QDP and PEQ intersect in S .

Show that the circumcircles of SDQ and SEP are tangent.

Solution



Note that S is the circumcenter of triangle PFQ . Indeed, by the SAS congruence criterion, $\triangle PDS \cong \triangle FDS$ and $\triangle PES \cong \triangle GES$, and in particular $SF = SP = SG$. Also,

$$\angle QFG = \angle DFB = \angle CBA = \angle ACB = \angle CGE = \angle FGQ,$$

so $FQ = GQ$ and we can now use the SSS congruence criterion to get $\triangle FQS \cong \triangle GQS$, and in particular that SQ is the angle bisector of $\angle GQF$.

We now know S to lie on the angle bisectors of $\angle PEQ$, $\angle EQD$, and $\angle QDP$, so $d(S, EP) = d(S, EQ) = d(S, QD) = d(S, DP)$ and S also lies on the angle bisector of $\angle DPE$.

Let T be a point on the tangent to the circumcircle of SDQ at S , so that $\angle TSD = \angle SQD$. Using that S lies on the four angle bisectors of quadrilateral $DPEQ$, we get

$$\begin{aligned} \angle PST &= \angle PSD - \angle TSD \\ &= 180^\circ - \angle SDP - \angle DPS - \angle SQD \\ &= \frac{1}{2}(360^\circ - \angle QDP - \angle DPE - \angle EQD) \\ &= \frac{1}{2}\angle PEQ \\ &= \angle PES, \end{aligned}$$

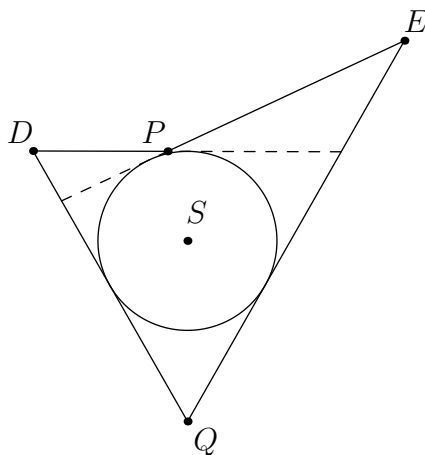
so T also lies on the tangent to the circumcircle of SEP at S . This proves the circumcircles of SDQ and SEP have the same tangent through S , i.e. they are tangent at S .

Solution 2: As in the previous solution, we show that $FQ = GQ$ by angle chasing. Then,

$$DP + EQ = DP + EG + GQ = DF + EP + FQ = DQ + EP$$

and by the converse of Pitot's theorem, the quadrilateral $DPEQ$ has an inscribed circle. In particular, its four angle bisectors are concurrent and their intersection is S , showing that both QS and PS are angle bisectors of $\angle EQP$ and $\angle DPE$, respectively. We can then conclude as in the previous solution.

Remark: The incircle of $DPEQ$ touches the extensions of the sides DP and PE , not the sides themselves.



Marking Scheme

(A) **5P: Showing that S is the incenter of $DPEQ$**

(A1) **1P:** Showing that $QF = QG$.

(A2) **1P:** Showing that $SF = SG$.

(A3) **4P:** Showing that quadrilateral $DPEQ$ has an inscribed circle.

(A4) **4P:** Showing that $\angle EQS = \angle SQD$ **or** $\angle DPS = \angle SPE$.

(A5) **5P:** Showing that $\angle EQS = \angle SQD$ **and** $\angle DPS = \angle SPE$.

(B) **2P: Concluding**

(B1) **1P:** Showing that it is enough to prove that $\angle PSD = \angle SQD + \angle PES$ or that $\angle QSE = \angle QDS + \angle SPE$.

Remark: (A1) and (A2) are additive.